

AI for Mathematics

Session 1: From Counting to AI for Mathematics

DSA5210 – National University of Singapore

13 May 2026

A Problem Knuth Had Been Working On



Donald Knuth (b. 1938)

American computer scientist at Stanford; author of *The Art of Computer Programming* and creator of the TeX typesetting system.

In **Claude's Cycles**, Knuth describes a graph-decomposition problem on a Cayley digraph, an AI-assisted search process, and a later human proof.

- The paper is dated 28 February 2026 and revised 14 April 2026.
- The AI-produced construction handled odd values of m .
- The even case is discussed separately in the postscript.

Sources: Knuth, "[Claude's Cycles](#)", 2026; Knuth [news page](#).

Claude's Cycles: What Happened

"Shock! Shock!"

-- Donald Knuth, "Claude's Cycles"

- Claude reformulated the problem in terms of a directed Cayley graph.
- The successful odd-m construction appeared after exploration 31.
- Knuth then wrote the mathematical proof.
- A public Lean 4 formalization later formalized the main theorem.

The episode records search, programs, a construction, a human proof, and a later Lean formalization.

Formalization link

[kim-em/KnuthClaudeLean](#)

- Lean 4 repository.
- Formalizes results from Knuth's note.
- Main theorem:
`ClaudesCycles.hamiltonian_arc_decomposition`.

This is a source link, not a diagram of an agent system.

Before Proof

Mathematics as record, measure, and calculation

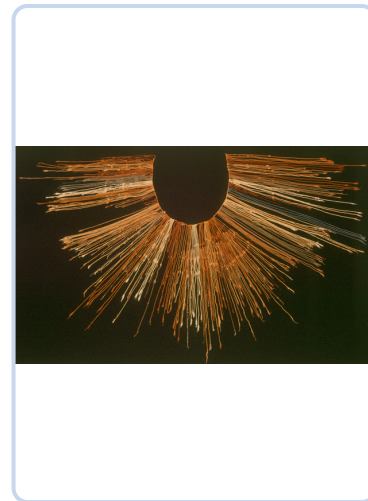
Counting Records

- The oldest mathematical traces are records: marks, knots, tokens, quantities.
- The Ishango bone is often discussed as an early tally artifact.
- Quipu records used knots and positions to encode information in the Andes.

At this stage mathematics is tied to memory: what has been counted, owned, owed, measured, or scheduled.



Tally artifact



Quipu

Calculating Shape

- Ancient calculation handled land, building, astronomy, accounting, and calendars.
- Plimpton 322 is a Babylonian tablet associated with Pythagorean triples.
- Such artifacts show structured computation before the Greek proof tradition.

$$3^2 + 4^2 = 5^2$$

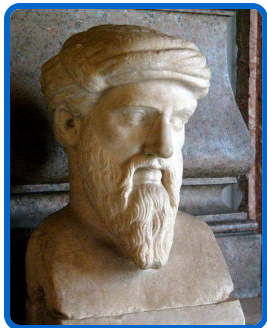


Babylonian mathematical tablet associated with Pythagorean triples.

Proof Appears

From examples to reasons

Pythagoras and $\sqrt{2}$



Pythagoras (c. 570-495 BCE)

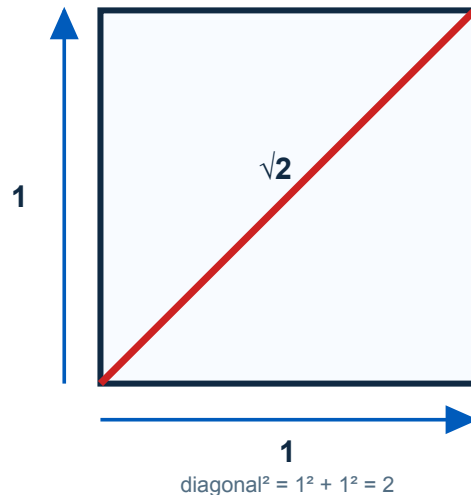
Greek philosopher and mathematician associated with the Pythagorean school and the Pythagorean theorem.

$\sqrt{2}$ is irrational.

Hippasus legend: later sources connect disclosure of irrationality with drowning; this is not treated as established fact.

Sources: Pythagoras; square root of 2; SEP Pythagoreanism; Britannica Hippasus.

Unit Square



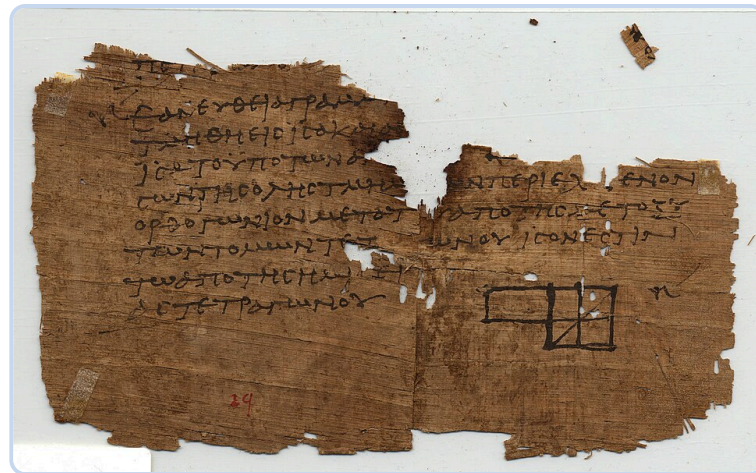
A unit square has diagonal length $\sqrt{2}$.

Euclid's *Elements*



Euclid of Alexandria (active c. 300 BCE)
Greek mathematician associated with *Elements*, a systematic presentation of geometry and number theory.

- Definitions, postulates, common notions.
- Propositions arranged so later claims depend on earlier proofs.
- A diagram may guide the eye; the proof carries the conclusion.



Fragment of Euclid's *Elements*.

Storing Mathematical Knowledge



Al-Khwarizmi (c. 780-850)

Mathematician and astronomer in Baghdad; his works are tied to the words algebra and algorithm.

9th c.

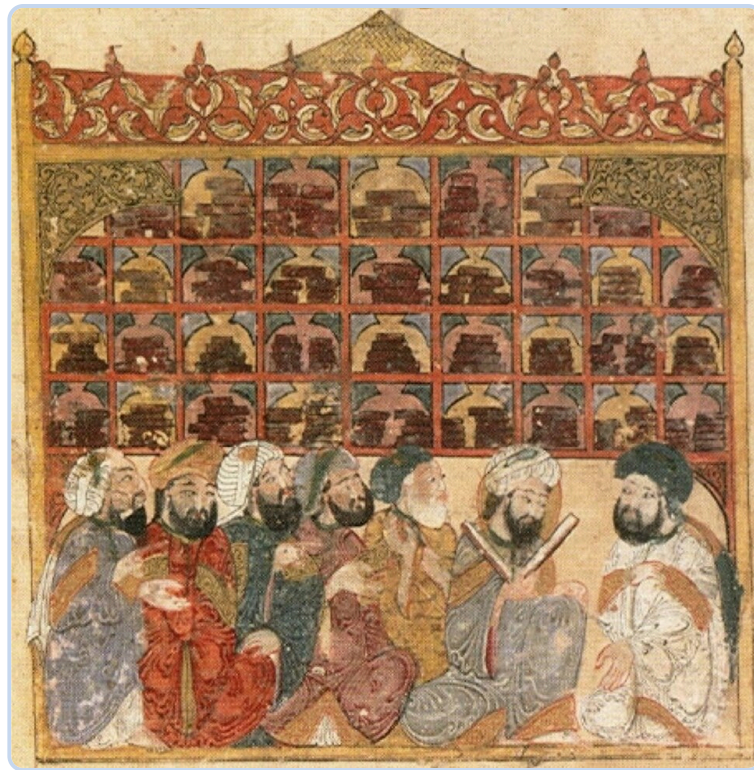
Translation and study around Baghdad's House of Wisdom.

2005-

Stacks Project as an open, linkable reference for algebraic geometry.

2017-

Mathlib as a community library for formalized mathematics in Lean.



The House of Wisdom in Baghdad, represented in a historical illustration.

Proofs Become Hard To Check

Length, specialization, and verification

Fermat's Last Theorem: Statement and Gap



Pierre de Fermat (1607-1665)

French lawyer and mathematician whose margin note led to Fermat's Last Theorem.



Andrew Wiles (b. 1953)

British mathematician who proved Fermat's Last Theorem in the 1990s.

$a^n + b^n = c^n$, with $n > 2$, has no positive integer solutions.

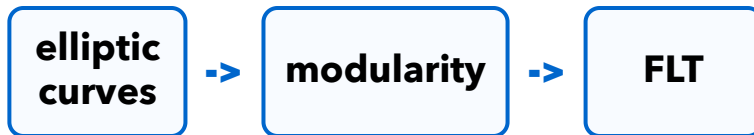
Fermat wrote his claim in 1637. Wiles announced a proof in 1993; after a gap was found, a corrected proof appeared in 1994 and was published in 1995.

Andrew Wiles: A Modern Proof



Andrew Wiles proved a deep case of the modularity theorem for elliptic curves, from which Fermat's Last Theorem follows.

- Modern proof can depend on a large body of specialized theory.
- Checking is distributed across experts and texts.
- A formal project must also formalize prerequisites.



This is the kind of proof where the statement is short but the verification context is large.

ABC Conjecture: Verification as Part of the Story



Shinichi Mochizuki (b. 1969)

Japanese mathematician at Kyoto University's RIMS, known for work in arithmetic geometry and Inter-universal Teichmüller theory.

- In 2012 he released papers claiming a proof of the ABC conjecture.
- The claimed proof uses Inter-universal Teichmüller theory.
- Acceptance and verification have been debated among experts.

For this course, ABC is a case where the verification process itself became part of the mathematical story.

Sources: Asahi/Kyoto University photo caption, 2021 article; Plus Magazine overview; ABC conjecture.

ABC: The Scholze-Stix Objection



Peter Scholze (b. 1987)

German arithmetic geometer and Fields Medalist; known for perfectoid spaces.



Jakob Stix (b. 1974)

German arithmetic geometer at Goethe University Frankfurt; works in anabelian geometry.

- In 2018 Scholze and Stix wrote *Why abc is still a conjecture*.
- Their objection focuses on the step around IUT III, Corollary 3.12.
- The Mochizuki side disputed their interpretation and whether it constitutes a gap.

This is an example of mathematical verification becoming a public, technical disagreement.

Sources: Peter Scholze and Jakob Stix, *Why abc is still a conjecture*, 2018; Quanta, Titans of Mathematics Clash Over Epic Proof of ABC Conjecture, 2018; Stix photo: Wikimedia Commons.

LANA: Lean Formalization of IUT

LANA stands for **Lean for ANabelian geometry**.

- The project digitizes definitions and results from anabelian geometry using Lean4.
- ZMC announced LANA on 31 March 2026.
- The project says preparation began in autumn 2023 and full operation began in September 2024.

One aim is to formalize major theorems of anabelian geometry and build a Lean library.

Another aim is to verify Inter-universal Teichmüller theory through Lean formalization, from a neutral standpoint.

The target is not an AI opinion about ABC; it is a proof-assistant route to precise statements and checkable dependencies.

Formal Foundations and Proof Checking

Can checking become mechanical?

Hilbert's Program



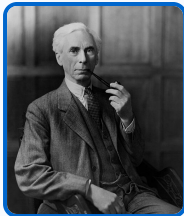
David Hilbert (1862-1943)

German mathematician whose work shaped algebra, geometry, analysis, foundations, and mathematical physics.

Hilbert's program aimed to place mathematics on explicit formal foundations and prove their consistency using finitary methods.

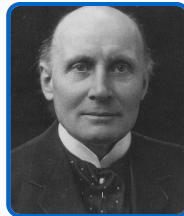
- Mathematical Problems (1902): the list of 23 Hilbert problems.
- Formal language.
- Explicit proof rules.
- Mechanical checking as an imagined endpoint.

Principia Mathematica



**Bertrand
Russell** (1872-
1970)

British
philosopher,
logician, and
mathematician.



**Alfred N.
Whitehead**
(1861-1947)

British
mathematician
and
philosopher.

Principia Mathematica (1910-1913) attempted to derive mathematics from logic through explicit symbolic rules.

PRINCIPIA MATHEMATICA

TO *56

BY
ALFRED NORTH WHITEHEAD
AND
BERTRAND RUSSELL, F.R.S.



CAMBRIDGE
AT THE UNIVERSITY PRESS

The famous $1 + 1 = 2$ result appears hundreds of pages into the work.

Gödel's Incompleteness Theorems



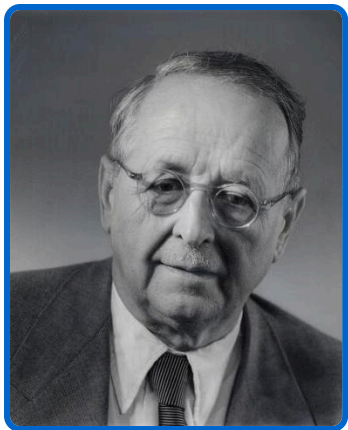
Kurt Gödel (1906-1978)

Austrian-American logician whose incompleteness theorems changed the foundations of mathematics.

For any consistent, effectively axiomatized formal system T that can express elementary arithmetic:

1. There is an arithmetic sentence φ_T such that T proves neither φ_T nor its negation $\neg\varphi_T$.
 2. T cannot prove its own consistency statement $\text{Con}(T)$.
- Gödel's incompleteness theorems limit Hilbert's hope for one complete formal foundation.
 - They do not prevent formal checking of particular proofs in a specified system.

Weyl: Computing Machines as Outside Help



Hermann Weyl (1885-1955)

German-American mathematician; worked in analysis, geometry, mathematical physics, topology, and foundations.

In 1946, Weyl publicly raised the possibility that high-speed computing machines could become outside help for increasingly difficult mathematical work.

- This is before modern proof assistants.
- The following examples involve search, checking, and computer-assisted calculation.

Sources: Hermann Weyl; Princeton Companion to Mathematics, article VI.80.

Automated Proof Search: Logic Theorist

1956

Logic Theorist searches for proofs in symbolic logic.

Input

Theorems from Chapter 2 of *Principia Mathematica*.

Result

Proofs for 38 of the first 52 target theorems.

This is the proof-search tradition: the machine tries to find a derivation from formal rules.

It is different from proof checking: a found proof still needs a trusted way to verify each step.

Source: History of Information, Logic Theorist.

Proof Checking: Automath

1968

Automath is designed as a language for fully written mathematical proofs.

Object

A proof becomes a precise formal object, not only a prose argument.

Checker

The computer verifies whether the proof object follows the rules.

Automath makes the checking problem explicit: if a proof is written in the formal language, a small checker can inspect it line by line.

```
ε * N := PN : set
ε * 0 := PN : N
ε * n := --- : N
n * s := PN : N
ε * 1 := s(0): N
```

--- opens an assumption line. PN introduces a primitive notion. A definition line gives the body and the type.

Computer-Assisted Proof: Four Colors

1976

The Four Color Theorem is proved using computer-assisted case checking.

Issue

Part of the proof depends on computations too large to check by hand in the usual way.

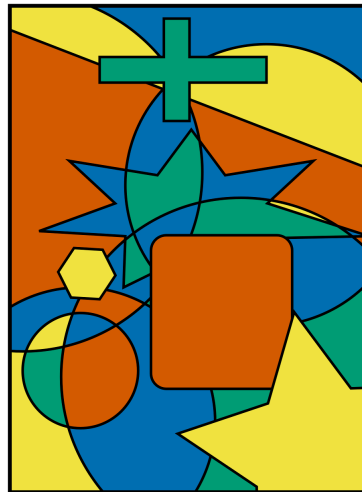
2005

A proof is formalized and mechanically checked in Coq.



Georges Gonthier

Computer scientist; led the Coq formalization of the Four Color Theorem.



A planar map colored with four colors.

Sphere Packing and Flyspeck



Johannes Kepler (1571-1630)

German astronomer and mathematician; formulated the sphere-packing conjecture in 1611.



Thomas Hales (b. 1958)

American mathematician; proved the Kepler conjecture and led the Flyspeck formalization.

1611

The conjecture asks for the densest packing of equal spheres in three-dimensional Euclidean space.

1998

Hales announced a computer-assisted proof.

2014

The Flyspeck project completed a formal proof using HOL Light and Isabelle.

Optimal density: $\pi/\sqrt{18}$.

Kepler Packing in One Picture



Equal balls packed in layers: a physical model for the Kepler conjecture.

Image: University of Michigan Library Digital Collections, Thomas Hales demonstrating his solution to the Kepler Conjecture, CC-BY-NC 4.0.

Proof Assistants Become Infrastructure

1970s

Mizar and HOL-family provers help establish reusable formal proof environments.

1980s

Coq and Isabelle develop proof languages, logics, automation, and library practices.

1994

The QED Manifesto proposes a computer-based body of formalized mathematical knowledge with mechanically checked proofs.

Library

Reusable definitions, libraries, notation, automation, and checkers become central.

Modern

Lean and Mathlib are a library-centered proof-assistant ecosystem.

Sources: QED Manifesto, 1994; "Twenty Years of the QED Manifesto", 2016.

Lean, Mathlib, Formal Libraries

Proofs as checkable objects

Why Lean Was Built

Earlier systems

- Automath, Mizar, Coq, Isabelle, HOL-family systems.
- Each explored a different balance of logic, automation, libraries, and interaction.

Lean's design space

- Dependent type theory with a small trusted kernel.
- Automation and tactic programming.
- A programming language for metaprogramming.
- A large community library for mathematics.

The question was not simply "Can a machine check proofs?" It was how to make formal mathematics practical enough for a growing community.

Leonardo de Moura and Lean



Leonardo de Moura

Computer scientist known for the Z3 SMT solver and as the creator of the Lean theorem prover.

- Lean began at Microsoft Research around 2013.
- Lean 4 is both a theorem prover and a programming language.
- The system supports formal proof, automation, and metaprogramming in one environment.

For AI-for-math, Lean provides a checker, a language, and a searchable mathematical codebase.

Sources: de Moura homepage; Lean official site; Microsoft Research Z3 project.

What Lean Changes About Trust

Traditional checking

Reader checks the written argument.

Trust depends on expertise, exposition, and social review.

Long prerequisites may be implicit.

Lean checking

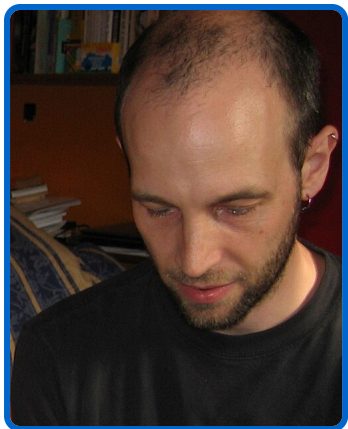
The kernel checks a formal proof term.

Checking is independent of the prover's identity.

The formal statement must match the intended theorem.

Careful wording: Lean does not make every written proof automatically correct. It checks a formalized theorem and proof inside a specified environment.

Kevin Buzzard, Xena, and Mathlib



Kevin Buzzard

British number theorist at Imperial College London; an advocate for formalizing modern and undergraduate mathematics in Lean.

- The Xena Project used Lean to formalize undergraduate mathematics.
- Undergraduate curricula give controlled tests for definitions, APIs, and missing lemmas.
- Reusable results can become part of Mathlib.

Mathlib did not simply start as one undergraduate project. But projects such as Xena helped expose what a real mathematical library needs.

What Mathlib Is

- Mathlib is Lean's community library of formalized mathematics.
- It contains definitions, named theorems and lemmas, tactics, notation, typeclass conventions, documentation, and examples.
- A theorem in Mathlib is a named declaration in a checked environment.
- Reusable mathematics can be called by name instead of reproved from scratch.

```
import Mathlib

#check Nat.Prime
-- Nat.Prime : Nat → Prop

#check Nat.Prime.dvd_mul
-- Nat.Prime.dvd_mul {p m n : ℕ} (pp : Nat.Prime p) :
--   p ∣ m * n ↔ p ∣ m ∧ p ∣ n

example (p m n : Nat) (hp : Nat.Prime p) (h : p ∣ m * n) :
  p ∣ m ∧ p ∣ n := by
  exact (Nat.Prime.dvd_mul hp).mp h
```

The slide example uses a library predicate, a library theorem, and one checked proof that reuses the theorem.

Formalization Projects

2005

Four Color Theorem: Coq verification after the 1976 computer-assisted proof.

2003-2014

Flyspeck: a formal proof of Hales's computer-assisted Kepler-conjecture proof.

2020-2022

Liquid Tensor Experiment: modern condensed/liquid mathematics in Lean.

2023

PFR formalization: public Lean development around additive combinatorics.

2024-2029

FLT Project: ongoing Lean formalization of Wiles's proof route.

2025

Strong PNT / Gauss: AI-assisted Lean formalization report.

2026

Viazovska sphere packing: reports on Lean verification for the 8- and 24-dimensional cases.

2026

Knuth's Claude cycles proof: focused Lean formalization of a recent AI-assisted episode.

A project case file asks: start date, target theorem, formal material built, story, and public artifact.

Formalization Time Scales

2005

Four Color: Gonthier's Coq formalization; XMUM table records about six years.

2014

Kepler / Flyspeck: formal proof completed after an eleven-year project.

2022

Liquid Tensor: main theorem announced after an intensive Lean effort.

2023

PFR: initial theorem formalized in weeks through a public blueprint/repository workflow.

2024-29

FLT Project: funded period for an ongoing public Lean formalization project.

2025

Strong PNT: Math Inc. reports a September 2025 Lean formalization using Gauss.

2026

Viazovska sphere packing: project history begins in 2024; public reports continue in 2026.

These are artifact timelines, not race times. Scope, library state, automation, and human coordination differ.

Liquid Tensor Experiment



Peter Scholze (b. 1987)

German mathematician, Fields Medalist, known for perfectoid spaces and arithmetic geometry.

- Challenge posed in December 2020.
- Main Lean theorem announced in July 2022.
- Target came from condensed/liquid mathematics.

Sources: [Liquid Tensor Experiment](#); [Lean community completion announcement](#); [Peter Scholze](#).

$$\mathrm{Ext}^i(\mathcal{M}_{p'}(S), \mathcal{M}_p(S)) = 0 \quad (i > 0)$$

- The project used a public blueprint and Lean files.
- It built infrastructure around normed groups, power series, homological algebra, measures, and condensed abelian groups.

PFR Formalization



Terence Tao (b. 1975)

Australian-American mathematician, Fields Medalist, known for work across analysis, number theory, combinatorics, and mathematical exposition.

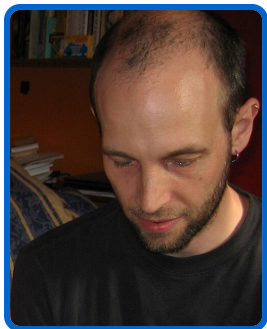
- Proof target released in 2023.
- Lean formalization organized through a public blueprint, documentation page, and repository.
- The initial theorem was formalized in weeks.

Source: [Tao's PFR formalization project.](#)

If $A \subset \mathbb{F}_2^n$ and $|A + A| \leq K|A|$, then A is covered by at most $2K^{12}$ cosets of a subspace H with $|H| \leq |A|$.

The work includes additive-combinatorics definitions, finite-vector-space structure, cosets, subspaces, and entropy inequalities.

FLT Project



Kevin Buzzard's FLT Project aims to formalize the proof of Fermat's Last Theorem in Lean.

- Ongoing public Lean project; UKRI/EPSRC funding period is September 2024 to September 2029.
- Target: a Lean formalization of Wiles's proof route.
- Prerequisites include elliptic curves, modular forms, Galois representations, modularity, and surrounding number theory.

$$a^n + b^n = c^n, \quad n > 2$$

The theorem statement is short. The formalization task is long because the proof route depends on a large modern number-theory library.

Source: Imperial College London, FLT Project.

Math Inc. Gauss and Strong PNT



Carl Friedrich Gauss (1777-1855)

German mathematician whose name is used here for Math Inc.'s AI agent, not as a historical claim about the theorem.

$$\pi(x) \sim x / \log x$$

- Strong PNT is presented as a September 2025 Lean formalization project.
- The named system is Math Inc.'s Gauss.
- The project page reports about 25,000 lines of Lean and about 1,100 theorem/definition items.
- The artifact to inspect is the Lean development and its supporting infrastructure.

Sources: Strong PNT project page; Math Inc. Gauss; XMUM GE week 1 references.

Viazovska Sphere Packing Verification



Maryna Viazovska (b. 1984)
Ukrainian mathematician and Fields Medalist known for solving the sphere-packing problem in dimension 8, with related work in dimension 24.

dimension 8: E_8 lattice
dimension 24: Leech lattice

Sources: IEEE Spectrum, AI proof verification; Maryna Viazovska; sphere packing in dimension 8 and dimension 24; XMUM GE week 1 references.

2016

Viazovska proves optimality of the E_8 lattice packing in dimension 8.

2024

IEEE Spectrum reports that the AI-assisted formalization effort began in March 2024.

2025

Public repository access is reported in June 2025.

2026

Reported milestones include the dimension 8 case and later the dimension 24 Leech-lattice case.

Reported artifacts include 200,000+ lines of Lean for the dimension 24 case and a typo found during the process.

AI Enters Mathematical Reasoning

Generation, search, tools, and verification

IMO Benchmarks: Solution Form and Verification

Start with IMO because the scoring object is concrete: six problems, seven points each, 42 points total.

- What problem set was used?
- What form did the proposed solution take?
- How was that proposed solution verified?
- Was natural language translated into a formal representation?

Solution form first

- Natural-language proof.
- Formal proof file.
- Geometry construction plus symbolic checker.
- Hybrid route through translation.

Research-frontier benchmarks come later, after the IMO cases.

Sources: [IMO Grand Challenge](#); [Google DeepMind, IMO 2024 post](#); [Google DeepMind, IMO 2025 post](#).

IMO Grand Challenge

- The IMO Grand Challenge proposed a benchmark for AI systems solving International Mathematical Olympiad problems.
- Each IMO problem is worth 7 points, for a total of 42.
- Modern systems differ in solution form and verification route.

Read each announcement by asking: What kind of solution is being claimed, and how was it verified?

Solution form	Verification route
Natural-language proof	Human grading or expert review
Formal proof file	Lean kernel / typechecking
Geometry construction	Symbolic geometry engine
Hybrid workflow	Translation plus one of the above routes

These are different routes to mathematical trust, not a ranking.

IMO 2024: Formal and Neuro-Symbolic Route

Jan 2024

AlphaGeometry combines a language model that proposes auxiliary constructions with a symbolic geometry engine.

Jul 2024

AlphaProof + AlphaGeometry 2 solve IMO problems at silver-medal standard.

Output

AlphaProof works with formal proof code; AlphaGeometry uses a symbolic checker in the geometry component.

Interface

Problems and proofs require translation between natural language and formal representations.

Source: Google DeepMind, "AI achieves silver-medal standard solving International Mathematical Olympiad problems", 2024.

AlphaProof: Lean Output Fragment

- DeepMind released Lean files for IMO 2024 problems P1, P2, and P6.
- The README says [originals/](#) contains direct agent output with explanatory comments added.
- Edited versions compile on a public Lean/mathlib release.

```
-- IMO 2024 P1 excerpt, not a standalone file
existsλx L⇒(L 2 two_pos).rec λl Y⇒?_
useλy . x⇒y.rec λS p⇒?_
· simp_all[λL:ℕ⇒(by norm_num[Int.floor_eq_iff]:
  [(L:ℝ)*S]=L*S)]
  rw[p.2, Int.dvd_iff_emod_eq_zero,
    Nat.lt_iff_add_one_le, ←Finset.sum_mul,
    ←Nat.cast_sum, S.even_iff] at *
  simp_all[Finset.sum_range_id]
```

The excerpt contains anonymous variables, placeholder goals, and dense tactic chains.

Source: Google DeepMind, [AlphaProof outputs from the 2024 IMO](#), public Lean package.

IMO 2025: Natural-Language Route

OpenAI announcement

- Experimental reasoning LLM.
- Natural-language proofs.
- Reported score: 35/42.
- Publicly announced by OpenAI researchers; not an official IMO entry.

Source: [Techmeme archive of the OpenAI announcement](#).

Gemini Deep Think

- Advanced Gemini model with Deep Think.
- End-to-end natural-language proofs from official problem descriptions.
- Officially achieved gold-medal standard.
- Google describes parallel, extended reasoning behavior.

Source: [Google DeepMind, official IMO 2025 post](#).

IMO 2025: Formal-Proof Route

Harmonic Aristotle

- Lean-oriented automated theorem-proving system.
- Reports IMO-level automated proving with formal proofs.
- Architecture emphasizes decomposition, search, and verification.

Source: Harmonic Team, "[Aristotle: IMO-level Automated Theorem Proving](#)".

Seed-Prover / Seed-Prover 1.5

- Lean proof generation with agentic reinforcement learning and tool use.
- Seed-Prover 1.0 participated in IMO 2025.
- Seed-Prover 1.5 later reports Lean proof code for five IMO 2025 problems.

Source: ByteDance Seed, [Seed-Prover 1.5 post](#); GitHub; [Seed-Prover paper](#).

AlphaProof and Seed-Prover Artifacts

AlphaProof public output

```
-- IMO 2024 P1 excerpt; not standalone
exists λx L⇒(L 2 two_pos).rec λl Y⇒?_
use λy . x⇒y.rec λS p⇒?_
· simp_all[λL:ℕ⇒(by norm_num[Int.floor_eq_iff]:
  [(L:ℝ)*S]=L*S)]
  rw[p.2, Int.dvd_iff_emod_eq_zero,
    Nat.lt_iff_add_one_le, ←Finset.sum_mul,
    ←Nat.cast_sum, S.even_iff] at *
  simp_all[Finset.sum_range_id]
```

Seed-Prover IMO 2025

```
-- IMO 2025 P5 excerpt; not standalone
lemma round1_main (l : ℝ)
  (hl : l > Real.sqrt 2 / 2):
  ∃ (N : ℕ),
    ((2 * (N : ℝ) + 1) * l
     - Real.sqrt 2 * (N : ℝ)) ^ 2 >
    2 * (N : ℝ) + 2 := by
  have h1 : 0 < Real.sqrt 2 := by
    exact Real.sqrt_pos.mpr (by norm_num)
  have h2 : l > 0 := by
    linarith [Real.sqrt_nonneg 2]
  -- proof continues in the source file
```

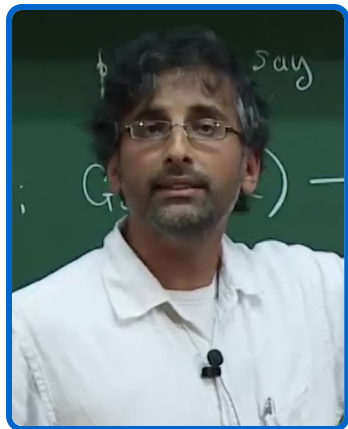
Both are proof artifacts. Seed-Prover additionally publishes English proof PDFs with claims and lemmas; that is the artifact-level sense in which it reads more like a human proof.

Sources: [Google DeepMind, AlphaProof IMO 2024 outputs](#); [ByteDance Seed, Seed-Prover GitHub repository](#); [ByteDance Seed, Seed-Prover 1.5 post](#).

Mathematicians on AI

Concrete public positions and working habits

Akshay Venkatesh



Akshay Venkatesh (b. 1981)

Australian mathematician, Fields Medalist, known for work connecting number theory, dynamics, topology, and representation theory.

- His public essays and interviews discuss how AI may change mathematical practice.
- One recurring theme is that AI can alter the roles of exploration, criticism, and exposition.
- For this course, his perspective belongs to the question of workflow, not only theorem proving.

Source: Institute for Advanced Study, ["The AI Revolution in Math Has Arrived"](#); [Quanta, 2026 interview/essay package.](#)

Timothy Gowers



Timothy Gowers (b. 1963)

British mathematician, Fields Medalist, known for functional analysis, combinatorics, and the Polymath collaborative projects.

- On 8 May 2026 he posted a detailed account of working with ChatGPT 5.5 Pro on a research problem.
- He describes the interaction as research collaboration rather than a finished formal verification pipeline.
- The post is useful here because it records a mathematician's working process with the model.

"PhD-level research in an hour or so" -- Gowers, 2026.

Terence Tao



Terence Tao appears twice in today's story: as a research mathematician using public collaboration, and as a public commentator on AI-math claims.

- His PFR formalization page shows a public, dependency-driven proof-development workflow.
- His commentary on AI-math announcements often distinguishes natural-language proof, formal proof, and human verification.
- That distinction is the same one used in the IMO architecture slides.

Sources: PFR formalization project; Tao's public blog.

Ken Ono, Axiom, and Lean Artifacts



Ken Ono

American number theorist known for modular forms, partitions, and arithmetic.



**Letong
(Carina) Hong**

Founder and CEO of Axiom Math; MIT graduate and Rhodes Scholar.

Axiom's public materials tie the system story to Lean, Mathlib, proof verification, and public proof artifacts.

- **AxiomProver:** theorem-proving system with public Lean artifacts.
- **AXLE:** Lean proof-verification and proof-manipulation API.
- **Putnam 2025:** repository says 8/12 solved by the end of the contest, remaining 4 in the following days.
- **Fel's Conjecture:** arXiv abstract says the proof is fully formalized in Lean/Mathlib and produced by AxiomProver from a natural-language statement.

From Benchmarks to Research Attempts

IMO

Fixed contest problems, public score, and proof artifacts that may be natural language or formal code.

2024

FrontierMath is introduced; its paper reports state-of-the-art models solving under 2% at release.

2026

FrontierMath: Open Problems tests proposed solutions to open research problems with programmatic verifiers.

2026

First Proof asks for checkable proof attempts on ten domain-specific research-level problems.

Epoch's 11 May 2026 FrontierMath update says an AI-assisted review flagged fatal errors in about one third of Tiers 1-4 problems.

Sources: Glazer et al., [FrontierMath, 2024](#); Epoch AI Tiers 1-4 update, [2026](#); Epoch AI Open Problems, [2026](#); OpenAI First Proof submissions, [2026](#).

Assessment

Assignments

Component	Weight
Assignment 1: AI-assisted natural-language proof	20%
Assignment 2: AI-assisted Lean formalization of A1	20%
Assignment 3: Group AI-for-math project	20%
Individual report: submitted by each student	40%

Assignment 3

- Teams of 3 students maintain a shared Git repository.
- Possible forms: agent design, theorem discovery, or mathematical problem solving.
- Examples: proof critique, Lean assistance, conjecture search, PDE problem solving.
- Individual grades may reflect contribution amount and quality.

Each artifact records the mathematical target, AI workflow, checking method, and failure analysis.

Open Problem for Live AI Dialogue

Hadwiger--Nelson problem

Color every point of $\mathbb{R}^2$ so that points at distance 1 have different colors.

What is the minimum number of colors?

Known range:

$$5 \leq \chi(\mathbb{R}^2) \leq 7.$$

Dialogue goals

- Ask AI whether the problem is solved.
- Ask for definitions: unit-distance graph, chromatic number, finite witness.
- Ask why a finite unit-distance graph gives a lower bound.
- Ask which claims are mathematical, computational, or still open.

Sources: [MathWorld](#), [Hadwiger-Nelson Problem](#); de Grey, 2018; [Polymath16](#) wiki.

After Class: Try an AI Coding Agent

Setup

- Register or verify a GitHub account.
- Access to an AI coding agent will be provided – details on Canvas.

What to try

- Ask it to analyze a Git repository: files, scripts, dependencies, build commands.
- Ask it to help with a math problem: assumptions, examples, possible proof routes.
- Ask it to review a proof draft: gaps, missing definitions, unsupported steps.
- For harder reasoning tasks, try `xhigh` mode and compare with a faster mode.

Keep a short note: what you asked, what it produced, and what you checked.